Volume III · Chapter 13 — Vector Databases & Knowledge Storage · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-13-context.md` for the AI interaction log.

Prerequisites: Ch 12.

Learning outcomes — the student can: explain embeddings storage and ANN search; use Chroma/FAISS (and know Weaviate/Pinecone); design metadata & filtering; reason about where knowledge lives in hardware (RAM/NVMe) and how it scales.

Labs (hands-on) — 12 h:

#	Lab	h
13a	Build & query a Chroma collection with metadata filters	3
13b	Compare FAISS index types (recall/speed)	3
13c	Persist & reload a vector store; measure memory footprint	3
13d	Hybrid search: combine BM25 + vectors	3

Datasets/tools: Chroma, FAISS, Weaviate (optional); embeddings model. **Assessment:** vector-store lab (**50%**); benchmark report (**30%**); quiz (**20%**). **Key decisions:** which vector DB; index type vs. recall/latency; RAM vs. disk; managed vs.

local. **References:** plan §13 → *RAG, vector databases & local inference*. **Hours:** Theory **8** + Lab **12** = **20**.